

Essential Terraform Commands for Cloud Automation Beginners

Terraform lets you create, update, and manage cloud servers and databases using simple code instead of clicking buttons in a cloud console. Learning these foundational commands will give you the confidence to deploy real cloud infrastructure safely and consistently.

What you'll learn:

1. Initialize Directory (terraform init)
2. Preview Before Building (terraform plan)
3. Deploy Resources (terraform apply)
4. Tear Down Lab Resources (terraform destroy)

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1. Initialize Directory (terraform init)

Before Terraform can do anything, it needs to download the required provider plugins (like AWS, Azure, or Google Cloud) into a hidden local folder. It also prepares your state storage backend so your team stays in sync. Run this command whenever you create a new configuration or add a new module. It is safe to run multiple times whenever you update provider versions.

```
$# Initialize workspace and download provider plugins  
terraform init
```

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2. Preview Before Building (terraform plan)

Terraform inspects your code, reads your existing cloud setup, and compares the differences. It then generates a detailed preview showing items marked with '+' (create), '~' (modify), or '-' (destroy). This dry run prevents accidental deletions and helps you spot unintended changes before any real infrastructure is touched.

```
$# Generate and view execution plan
terraform plan

# Save plan to a file for safe, reproducible apply
terraform plan -out=tfplan
```

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3. Deploy Resources (terraform apply)

This command executes the actions proposed in your configuration files to reach your desired infrastructure state. It will ask for an interactive 'yes' confirmation before making any real cloud changes. Terraform builds resources in the correct dependency order automatically, waiting for networks before creating virtual machines.

```
$# Apply configuration with interactive confirmation
terraform apply

# Or apply a previously saved plan directly
terraform apply tfplan
```

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4. Tear Down Lab Resources (terraform destroy)

When a project ends or you finish a weekend practice lab, this command deletes all cloud resources tracked in your state file in reverse order. It asks for confirmation before deleting anything. Running this ensures you never leave expensive test servers or load balancers running when you are done.

```
$# Destroy all managed infrastructure in the current project  
terraform destroy
```


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Key Takeaways

- + Always run terraform fmt to keep team code neat and standardized.
- + Review the '+' and '-' symbols in terraform plan before approving deployments.
- + Store your state file in remote storage like AWS S3 with state locking enabled.
- + Break large architectures into smaller reusable Terraform modules.
- + Use variable files (.tfvars) to separate environment-specific values like staging and prod.

Watch Out For

- ! Modifying cloud resources manually in the web console, causing state drift.
- ! Hardcoding sensitive API keys or passwords directly inside .tf files.
- ! Deleting or manually editing the terraform.tfstate file.
- ! Using auto-approve flags (terraform apply -auto-approve) in production environments.

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